



TRANSCRIPT OF RECORDS

Sara Chopra

University of Hamburg

20148 Hamburg

Germany

Tel.: +49 40 42838-0

Fax: +49 40 42838-6594

**Faculty of Mathematics, Informatics and
Natural Sciences**

Desired degree: Master of Science
Physics

The student has completed the degree.

Last Name: Chopra

First Name: Sara

Date of Birth: 1 April 2001

Place of Birth: Delhi

Enrolled on: 1 October 2022

Student ID No.: 7597332

Number/Type	Name	Semester/Date	ECTS Credits	Grade
Physics (M.Sc.)			121	1,29
Advanced Master's Courses			49	
PHY-MV-BE-E26	Advanced particle physics*		8	1,3
ÜB	Exercises in advanced particle physics*	Winter 22/23		
VL	Advanced particle physics*	Winter 22/23		
Final module exam(s): Block exam				1,3
PHY-MV-BE-E17	Seminar: Physics and application of laser-plasma accelerators*		3	1,0
SEM	Seminar: Physics and application of laser-plasma accelerators*	Winter 23/24		
Final module exam(s): Final exam				1,0
PHY-MV-BE-T12	Phenomenology of physics beyond the standard model*		6	1,0
ÜB	Exercises in phenomenology of physics beyond the standard model*	Winter 23/24		
VL	Phenomenology of physics beyond the standard model*	Winter 23/24		
Exam	Final exam			1,0
PHY-MV-BE-T02	Physics of the standard model*		6	1,0
ÜB	Exercises in physics of the standard model*	Summer 23		
VL	Physics of the standard model*	Summer 23		
Final module exam(s): Final exam			6 Jul 2023	1,0
PHY-MV-BE-T04	Quantum field theory I		8	1,3
ÜB	Exercises in quantum field theory I*	Winter 22/23		
VL	Quantum field theory I*	Winter 22/23		
Final module exam(s): Block exam			29 Mar 2023	1,3
PHY-MV-FN-T25	Symmetry groups in physics*		8	1,0
ÜB	Exercises in symmetry groups in physics	Summer 23		
VL	Symmetry groups in physics	Summer 23		
Final module exam(s): Final exam			8 Aug 2023	1,0

Number/Type	Name	Semester/Date	ECTS Credits	Grade
PHY-MV-FN-E39	Methods in nanobiotechnology I*		7	1,7
PR	Practical: methods in nanobiotechnology I*	Winter 23/24		
ÜB	Exercises in methods in nanobiotechnology I*	Winter 23/24		
VL	Methods in nanobiotechnology I*	Winter 23/24		
Final module exam(s): Module exam				1,7
PHY-MV-BP-E05	Seminar on biomedical physics I*		3	2,7
SEM	Seminar on biomedical physics*	Winter 22/23		
Exam	Presentation and paper			2,7
Complementary Subject			12	
SPLV	German - "B1.2"	Summer 23	6	b
Exam	Block exam			1,7
SPLV	German Intensive Course - "B2.1"	Winter 23/24	6	b
Exam	Block exam			2,0
Research Phase			30	
PHY-MF-BE-EP	Introductory project (Accelerator physics and particle physics)		15	b
Final module exam(s): Final assignment				b
PHY-MF-BE-VPb	Preparatory project (Accelerator physics and particle physics)		15	b
Final module exam(s): Final assignment				b
Master thesis			30	
PHY-MF-MA	Master's thesis		30	1,3
Final Paper		8 Oct 2025		1,3
Master's thesis:	Sum Rules for the Extended Higgs Sector Models: A comparative analysis of 2HDM, GM and the Septet Model*			
Examiners:	Prof. Dr. Gudrid Moortgat-Pick, Prof. Dr. Sven Heinemeyer			
Oral exam		9 Oct 2025		1,3

Number/Type	Name	Semester/Date	ECTS Credits	Grade
Additional			0	
SPLV	German - "B1.1"	Winter 22/23	6	1,7
Exam	Block exam			1,7
Current grade			excellent	1,29

A minimum of 120 ECTS credits is required to successfully complete the program.

Hamburg, 20 January 2026

This document is valid without signature.

Note: This document includes only successfully completed courses and examinations.

Grading system per component:

1,0 / 1,3	=	excellent
1,7 / 2,0 / 2,3	=	good
2,7 / 3,0 / 3,3	=	satisfactory
3,7 / 4,0	=	sufficient
5,0	=	insufficient

Calculations for the final grade and for individual components are determined by departmental regulations.

* = in the original language

b = pass

inc. = module/course incomplete

e = successfully completed



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Universität Hamburg

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Deutschland

Tel.: +49 40 42838-0
Fax: +49 40 42838-6594

**Fakultät für Mathematik, Informatik und
Naturwissenschaften**

Angestrebter Abschluss: Master of Science
Physics

Das Studium gilt als bestanden.

Nachname: Chopra
Vorname: Sara

Geburtsdatum: 1. April 2001
Geburtsort: Delhi

Immatrikuliert seit: 1. Oktober 2022
Matrikelnummer: 7597332

Nummer/Typ	Name	Semester/Datum	ECTS Credits	Note
Physics (M.Sc.)			121	1,29
Advanced Master's Courses			49	
PHY-MV-BE-E26	Advanced particle physics		8	1,3
ÜB	Exercises in advanced particle physics	WiSe 22/23		
VL	Advanced particle physics	WiSe 22/23		
Modulabschlussprüfung(en): Blockprüfung				1,3
PHY-MV-BE-E17	Seminar: Physics and application of laser-plasma accelerators		3	1,0
SEM	Seminar: Physics and application of laser-plasma accelerators	WiSe 23/24		
Modulabschlussprüfung(en): Modulabschlussprüfung				1,0
PHY-MV-BE-T12	Phenomenology of physics beyond the standard model		6	1,0
ÜB	Exercises in phenomenology of physics beyond the standard model	WiSe 23/24		
VL	Phenomenology of physics beyond the standard model	WiSe 23/24		
Prüfung	Modulabschlussprüfung			1,0
PHY-MV-BE-T02	Physics of the standard model		6	1,0
ÜB	Exercises in physics of the standard model	SoSe 23		
VL	Physics of the standard model	SoSe 23		
Modulabschlussprüfung(en): Modulabschlussprüfung			06.07.2023	1,0
PHY-MV-BE-T04	Quantenfeldtheorie I		8	1,3
ÜB	Exercises in quantum field theory I	WiSe 22/23		
VL	Quantum field theory I	WiSe 22/23		
Modulabschlussprüfung(en): Modulabschlussprüfung			29.03.2023	1,3
PHY-MV-FN-T25	Symmetry groups in physics		8	1,0
ÜB	Übung zu Symmetriegruppen in der Physik	SoSe 23		
VL	Symmetriegruppen in der Physik	SoSe 23		
Modulabschlussprüfung(en): Modulabschlussprüfung			08.08.2023	1,0

Nummer/Typ	Name	Semester/Datum	ECTS Credits	Note
PHY-MV-FN-E39	Methods in nanobiotechnology I		7	1,7
PR	Practical: methods in nanobiotechnology I	WiSe 23/24		
ÜB	Exercises in methods in nanobiotechnology I	WiSe 23/24		
VL	Methods in nanobiotechnology I	WiSe 23/24		
Modulabschlussprüfung(en): Modulabschlussprüfung				1,7
PHY-MV-BP-E05	Seminar on biomedical physics I		3	2,7
SEM	Seminar on biomedical physics	WiSe 22/23		
Prüfung	Referat und Hausarbeit			2,7
Complementary Subject			12	
SPLV	DaF - "B1.2"	SoSe 23	6	b
Prüfung	Blockprüfung			1,7
SPLV	DaF-Intensivkurs - B2.1	WiSe 23/24	6	b
Prüfung	Blockprüfung			2,0
Research Phase			30	
PHY-MF-BE-EP	Einarbeitungsprojekt (Beschleuniger- und Elementarteilchenphysik)		15	b
Modulabschlussprüfung(en): Projektabschluss				b
PHY-MF-BE-VPb	Vorbereitungsprojekt (Beschleuniger- und Elementarteilchenphysik)		15	b
Modulabschlussprüfung(en): Projektabschluss				b
Master thesis			30	
PHY-MF-MA	Masterarbeit Physik		30	1,3
Abschlussarbeit		08.10.2025		1,3
Thema:	Sum Rules for the Extended Higgs Sector Models: A comparative analysis of 2HDM, GM and the Septet Model			
Gutachtende:	Prof. Dr. Gudrid Moortgat-Pick, Prof. Dr. Sven Heinemeyer			
Kolloquium		09.10.2025		1,3

Nummer/Typ	Name	Semester/Datum	ECTS Credits	Note
Zusätzliche Leistungen			0	
SPLV	DaF - "B1.1"	WiSe 22/23	6	1,7
Prüfung	Blockprüfung			1,7
Gleitende Gesamtnote		sehr gut		1,29

Es sind 120 ECTS-Credits zu erbringen.

Hamburg, 20. Januar 2026

Dieses Dokument ist ohne Unterschrift gültig.

Hinweis: Dieses Dokument enthält nur bestandene Prüfungsversuche.

Bewertungssystem Teilleistungen:

1,0 / 1,3	=	sehr gut
1,7 / 2,0 / 2,3	=	gut
2,7 / 3,0 / 3,3	=	befriedigend
3,7 / 4,0	=	ausreichend
5,0	=	ungenügend

Die Art der Berechnung zur Ermittlung der Gesamtnote und der Teilleistungsnote wird in den Fachspezifischen Bestimmungen des Faches geregelt.

b = bestanden

uv. = Die Studienleistung ist noch nicht vollständig abgeschlossen.

e = erfolgreich erbracht

PORTFOLIO

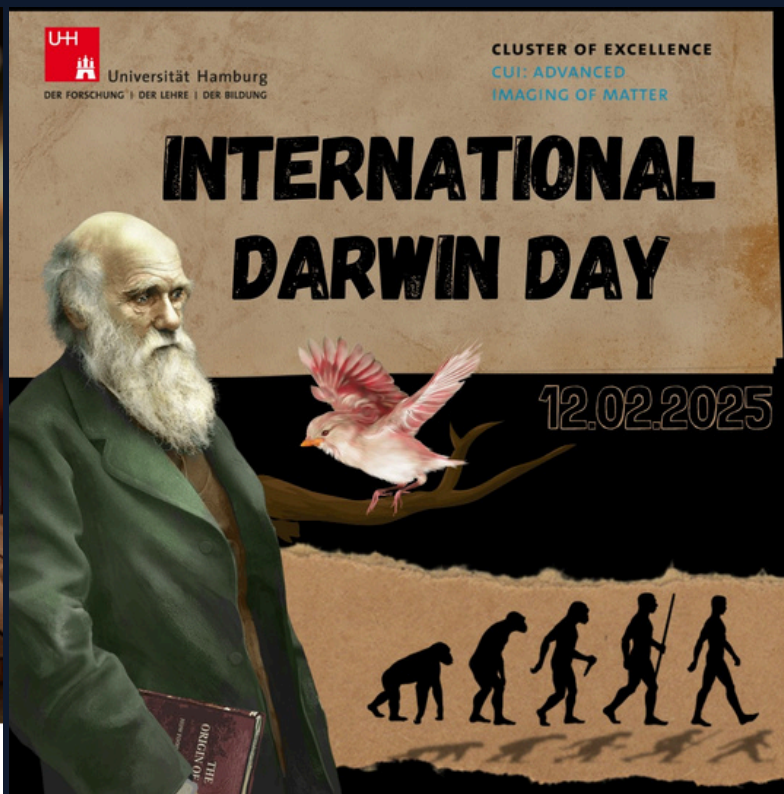
2026

Sara Chopra

CUI: Instagram Posts

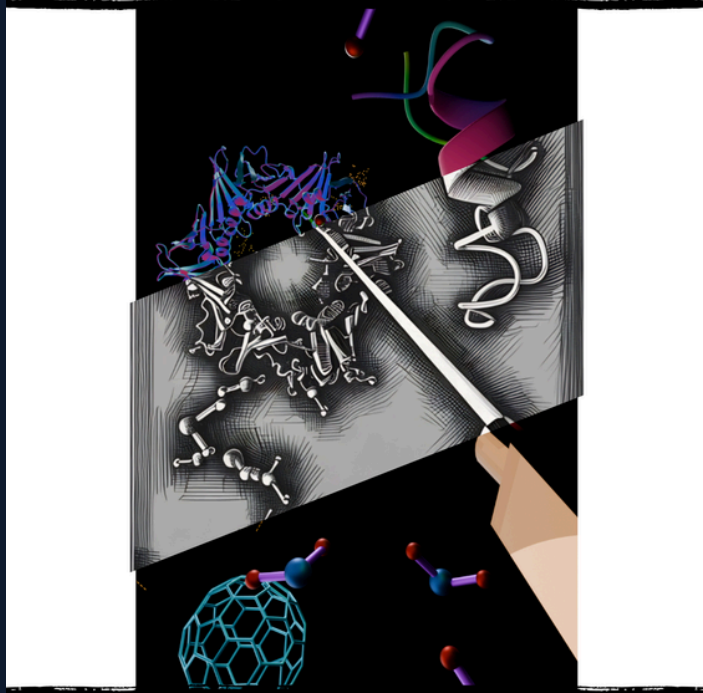


CUI: Instagram Posts



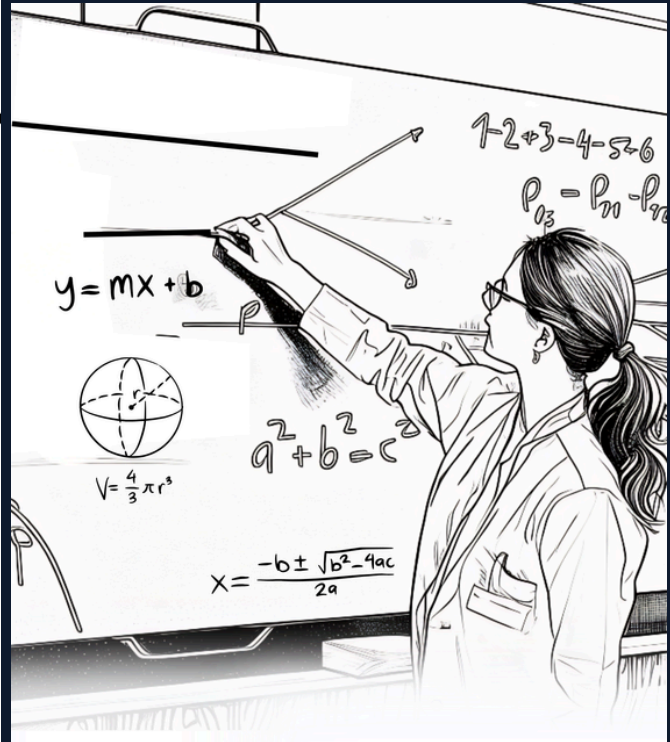
Other design samples

CUI: MALEBUCH



CLUSTER OF EXCELLENCE
CUI: ADVANCED
IMAGING OF MATTER

U+H Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG



„Dr. Maya erklärt, wie die Welt funktioniert, mit
Mathe, Physik und einer Tafel voller Ideen!“

“Dr. Maya explains how the world works, with math,
physics, and a whiteboard full of ideas!”

CUI: Research Slambook



CUI: ADVANCED
IMAGING OF MATTER

UNSERE FORSCHUNG
UNSERE MITARBEITER

U+H Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

CLUSTER OF EXCELLENCE
CUI: ADVANCED
IMAGING OF MATTER

CUI: SEITE DER GRUPPE

Gruppenfoto,
Illustration/Zeichnung
der Forschung

GROUP INFORMATION

PRINCIPLE
INVESTIGATOR _____
GRUPPENMITGLIEDER _____
HAUPTFORSCHUNGS
BEREICHE _____

WAS IST IHR DENKWÜRDIGSTER
CUI-MOMENT?

INTERESSANTE ERGEBNISSE/
WICHTIGE MEILENSTEINE

AKTUELLE FORSCHUNG
ZIELE UND METHODIK

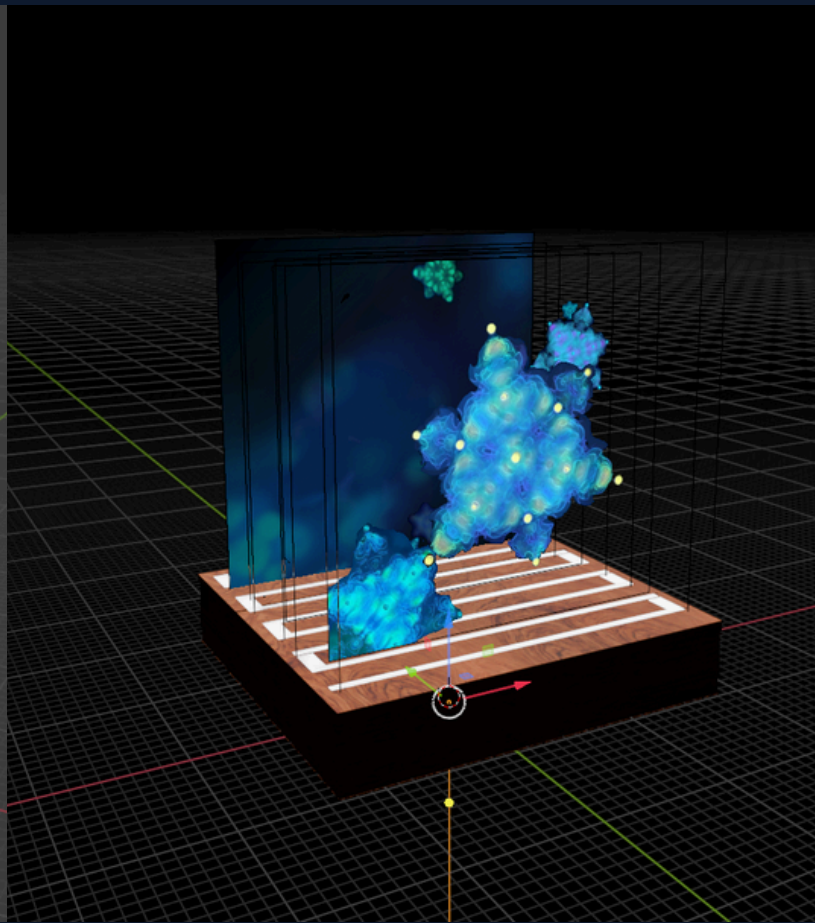
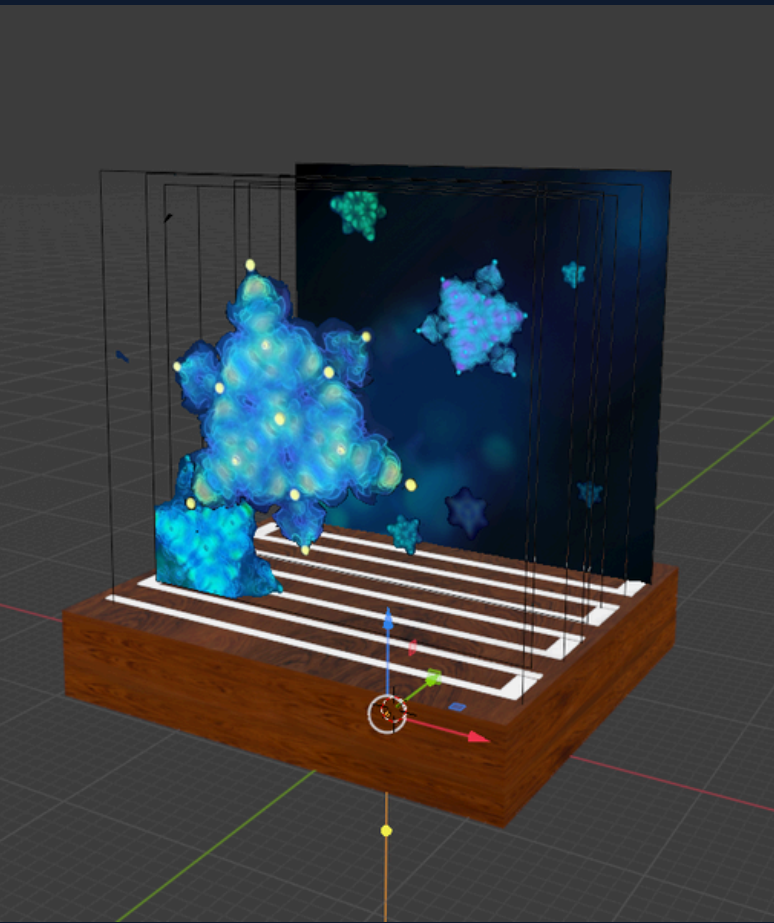
WAS IST DAS BESONDERE
AN IHRER ARBEIT?

WAS IST IHR CLUSTER-HIGHLIGHT
DER LETZTEN 7 JAHRE?

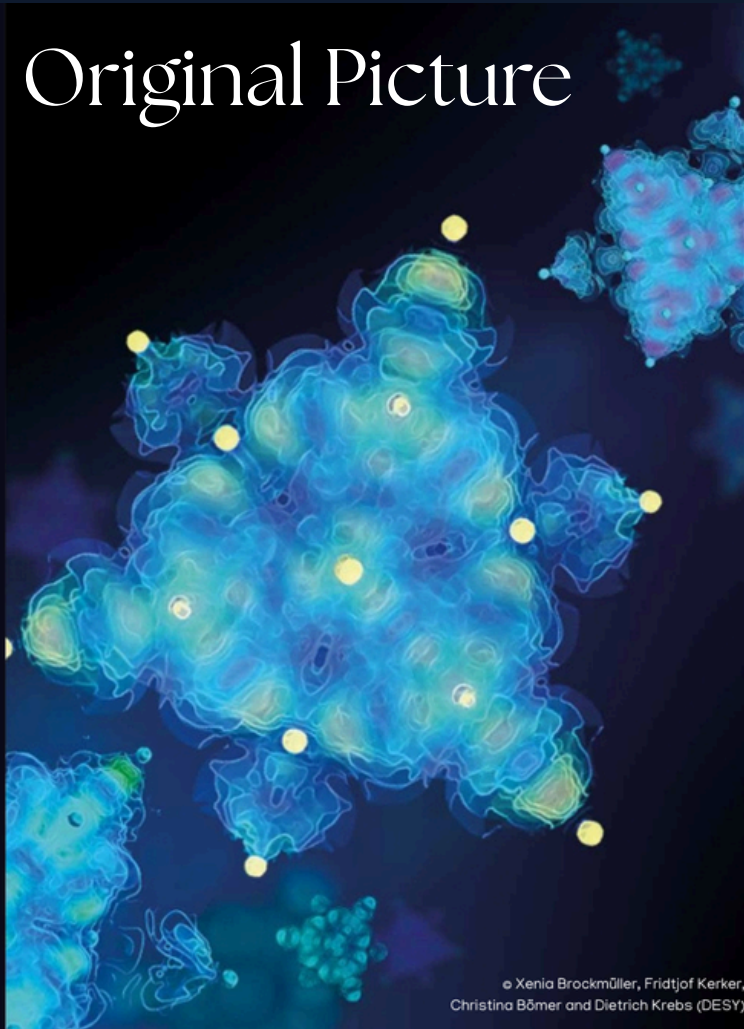
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CLUSTER OF EXCELLENCE
CUI: ADVANCED
IMAGING OF MATTER

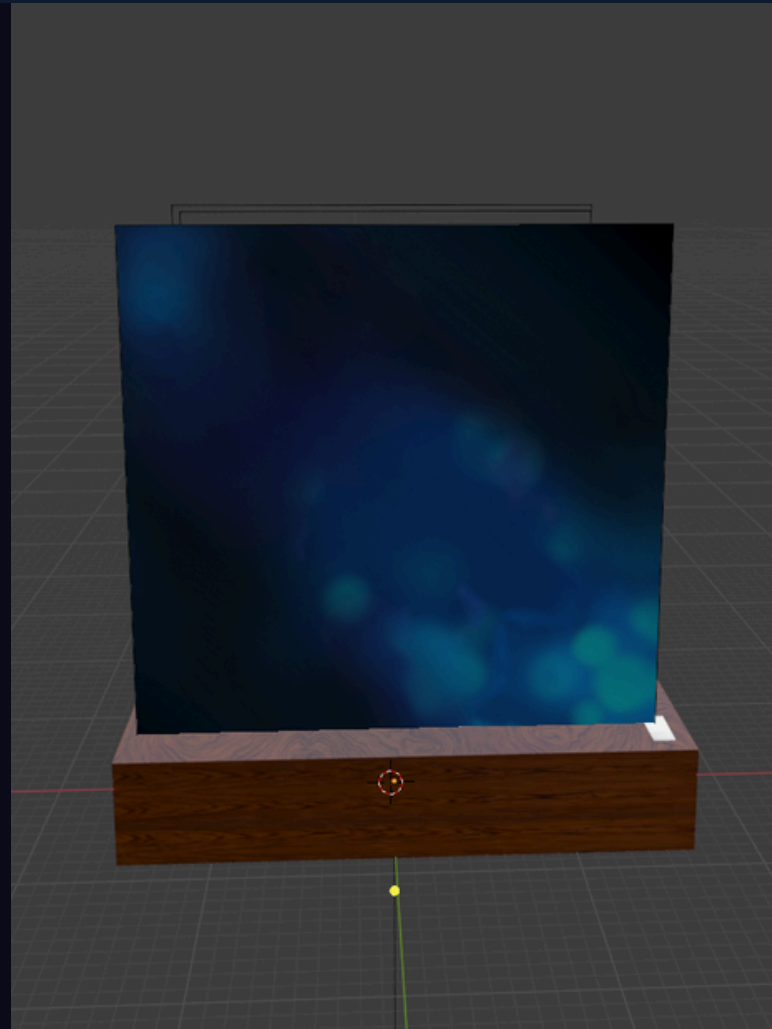
3D Model



Original Picture



© Xenia Brockmüller, Fridtjof Kerker,
Christina Bömer and Dietrich Krebs (DESY)



Physics Club (AIAS) Events



BEYOND GRAVITY

PHYSICS & Board Game

KUGELBLITZ Presents

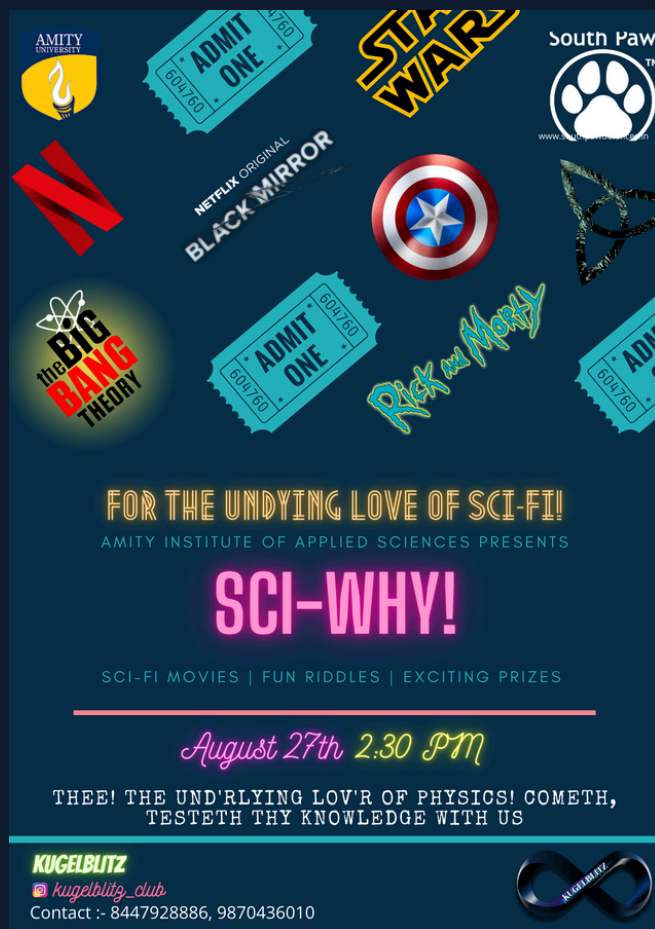
READY SET GO!!!

FLIPPITY

OCT 22
3:00 PM

@KUGELBLITZ_CLUB
CONTACT: 9870436010
8447928886

South Paw



AMITY UNIVERSITY

South Paw

ADMIT ONE

NETFLIX ORIGINAL **BLACK MIRROR**

STAR WARS

CAPTAIN AMERICA

the BIG BANG THEORY

Rick and Morty

ADMIT ONE

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FOR THE UNDYING LOVE OF SCI-FI!

AMITY INSTITUTE OF APPLIED SCIENCES PRESENTS

SCI-WHY!

SCI-FI MOVIES | FUN RIDDLES | EXCITING PRIZES

August 27th 2.30 PM

THEE! THE UND'RLYING LOV'R OF PHYSICS! COMETH,
TESTETH THY KNOWLEDGE WITH US

KUGELBLITZ
@kugelblitz_club
Contact :- 8447928886, 9870436010



AMITY UNIVERSITY

South Paw

AIAS & SOUTH PAW

KUGELBLITZ PRESENTS

PHYSICS QUEST

A HUNT FOR THE TREASURES OF PHYSICS

A QUEST THROUGH THE REALM OF PHYSICS

SEPTEMBER 24TH
3:00 PM

@KUGELBLITZ_CLUB

CONTACT: 9870436010, 8447928886

GirlUp Empower

The Truth About Geishas:

Artists or Prostitutes?



Mizuage (Virginity of Geisha)

A Geisha's Mizuage was sold by the Geisha House to Sponsor her expensive Training and Kimonos.



BURGER MENU

This is a
Burger!!



This is
not Meat!!



"HEME" is what makes these plant based meat,
FEEL like MEAT

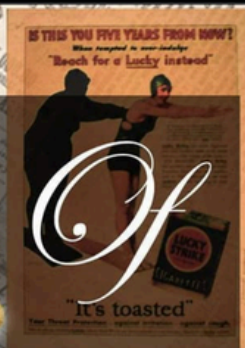


SIDES
CHEESE FRIES



IT IS EXTRACTED FROM SOYA PLANTS
& CAUSES THE EXPLOSION OF FLAVOUR & SIZZLE

Torches Of Freedom



MARLBORO
America's Finest Cigarette



It's toasted



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Smoking is a big problem these days, right? There are women, men and even children from all over the world who're addicted to smoking. Since smoking by women has been historically disapproved of, how did they get into it? To understand this, we have to go back to the first World War, when gender roles were diluted due to necessity. With men away at war, women were forced to move out of their homes and enter the workplace. Feminism became huge at this time. Women were demanding equal



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